

Storage in the Cloud TASK – 4

- 1. Create a table comparing Amazon EBS and Amazon S3, listing at least 4 key differences (such as data type, access method, use case, and scalability) and give one real-world app example where each would be the better choice.**

Ans :

Feature	Amazon EBS (Elastic Block Store)	Amazon S3 (Simple Storage Service)
Data Type	Block storage	Object storage
Access Method	Attached as a disk volume to an EC2 instance	Accessed through HTTP/HTTPS APIs, AWS SDKs, or the AWS Console
Primary Use Case	Operating systems, databases, boot volumes, and applications requiring low-latency storage	Storing images, videos, backups, documents, logs, and static website content
Scalability	Capacity can be increased, but volumes are attached to specific EC2 instances	Virtually unlimited storage with automatic scaling

Feature	Amazon EBS (Elastic Block Store)	Amazon S3 (Simple Storage Service)
Performance	High-performance, low-latency storage for frequent read/write operations	Optimized for durable storage and large-scale object access
Real-World App Example	Banking or MySQL database server – Uses EBS for fast database storage and reliable disk performance	Zomato image storage – Uses S3 to store restaurant images, menus, and user-uploaded photos because it is highly scalable and cost-effective

**2. Launch a new EC2 instance in the AWS Free Tier and attach a new EBS volume of at least 2 GB to it using the AWS Management Console.

Hint: After launching the instance, use the 'Volumes' section in EC2 to create and attach the volume.**

Ans :

- Sign in to the AWS Management Console.
- Open the Amazon EC2 service.
- Click Launch Instance.
- Enter a name for the instance (e.g., My-EC2-Instance).
- Select the Amazon Linux 2023 AMI (or another Free Tier eligible AMI).

- Choose the t2.micro or t3.micro instance type (Free Tier eligible, depending on your account).
- Create a new key pair or select an existing one.
- Configure the network settings (allow SSH (port 22) if you plan to connect to the instance).
- Leave the remaining settings as default and click Launch Instance.
- Wait until the instance state changes to Running.
- In the EC2 console, select Elastic Block Store → Volumes.
- Click Create Volume.
- Set the following:
 - Volume Type: General Purpose SSD (gp3)
 - Size: 2 GiB
 - Availability Zone: Select the same Availability Zone as your EC2 instance.
- Click Create Volume.
- Select the newly created volume.
- Click Actions → Attach Volume.
- Select your EC2 instance.
- Specify the device name (e.g., /dev/sdf or leave the default).
- Click Attach Volume.
- Verify that the volume status changes to In-use and is attached to your EC2 instance.

**3. Log in to your EC2 instance using SSH and format the attached EBS volume, then mount it to a new directory named /mnt/mydata.

Hint: Use lsblk to find your volume, mkfs to format, and mount to attach it to the directory.**

Ans :

- ssh -i your-key.pem ec2-user@<EC2-Public-IP>
- lsblk
- sudo mkfs -t ext4 /dev/xvdf
- sudo mkdir -p /mnt/mydata
- sudo mount /dev/xvdf /mnt/mydata
- df -h
- ls /mnt/mydata

4. Simulate a Zomato-style image upload: create a text file named zomato_menu.txt on your EC2 instance, copy it to your mounted EBS volume, and verify it is accessible from the new mount point.

Ans :

- ssh -i your-key.pem ec2-user@<EC2-Public-IP>
- echo "Zomato Menu Items: Pizza, Burger, Pasta, Biryani" > zomato_menu.txt
- ls -l zomato_menu.txt
- sudo cp zomato_menu.txt /mnt/mydata/
- ls -l /mnt/mydata/
- cat /mnt/mydata/zomato_menu.txt

5. Detach the EBS volume from your EC2 instance using the AWS Console, then reattach it to the same or a different instance. Describe what happens to the data you stored and why this behavior is useful for apps like Flipkart or Myntra order tracking.

Ans :

- Sign in to the AWS Management Console.
- Open the Amazon EC2 service.
- Go to Elastic Block Store → Volumes.
- Select the EBS volume attached to your EC2 instance.
- (Recommended) Stop the EC2 instance before detaching the volume:
- Select the instance.
- Click Instance State → Stop instance.
- Select the EBS volume.
- Click Actions → Detach Volume.
- Confirm the detachment.
- Wait until the volume status changes to Available.
- Launch another EC2 instance (or use the same stopped instance).
- Select the detached EBS volume.
- Click Actions → Attach Volume.
- Choose the target EC2 instance.
- Specify the device name (for example, /dev/sdf).
- Click Attach Volume.
- SSH into the EC2 instance.
- Mount the volume again:
- `sudo mount /dev/xvdf /mnt/mydata`
- What Happens to the Data?
- The data stored on the EBS volume remains intact even after detaching it from an EC2 instance.
- Detaching only removes the connection between the volume and the instance; it does not delete the stored files.

- When the volume is attached again to the same or another EC2 instance, the data can be accessed after mounting the volume.
- Why This Is Useful for Apps Like Flipkart or Myntra Order Tracking
- EBS provides persistent storage, which means application data can survive server changes or failures.